

Supplementary Information:

Stable and interchangeable deformation pathways preserve pelvic mechanical function across anatomical variability

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This Supplementary Information collects robustness, specificity, and model-boundary analyses that support the main manuscript. Throughout, the childbirth-related load cases remain linear proxy loads rather than nonlinear labour simulations.

Supplementary Note 1 — Spectral cluster architecture

The main manuscript focuses on the robust low-rank backbone and on a small number of recurrent clustered subspaces. The tables below provide the complete bookkeeping needed to track those clustered regimes across the cohort, including reference-mode occupancy, swap frequencies, and morphology-axis diagnostics.

Table S1 — Detected near-degenerate rank clusters

Cluster	Dim	n	Prev.	Med. gap _{in,min}	Med. sep _{out}	Type
Rank modes 9–10	2	152	54.7%	0.08114	0.4926	near-degenerate
Rank modes 12–15	4	116	41.7%	0.03153	0.4331	near-degenerate
Rank modes 5–6	2	94	33.8%	0.08737	0.2345	near-degenerate
Rank modes 12–13	2	78	28.1%	0.03839	0.1941	near-degenerate
Rank modes 14–15	2	70	25.2%	0.05116	0.1729	near-degenerate
Rank modes 12–14	3	45	16.2%	0.04109	0.1855	near-degenerate
Rank modes 4–5	2	35	12.6%	0.09521	0.2185	near-degenerate
Rank modes 13–15	3	29	10.4%	0.04465	0.1588	near-degenerate
Rank modes 7–8	2	21	7.6%	0.1051	0.3676	near-degenerate

Table S2 — Rank-cluster to reference-mode mapping

rank_cluster	dim	n_subjects	prevalence	paired_all_rate	ref_modes_mode	ref_modes_mode_rate	ref_modes_top3
Rank modes 9–10	2	152	54.7%	1	9–10	1	9–10 (100%)
Rank modes 12–15	4	116	41.7%	0.7414	12–15	1	12–15 (100%)
Rank modes 5–6	2	94	33.8%	1	5–6	0.8298	5–6 (83%); 4,6 (17%)
Rank modes 12–13	2	78	28.1%	0.9872	13–14	0.4805	13–14 (48%); 12–13 (44%); 12,15 (6%)
Rank modes 14–15	2	70	25.2%	0.8571	12,15	0.45	12,15 (45%); 14–15 (38%); 13–14 (15%)
Rank modes 12–14	3	45	16.2%	0.9333	12–14	0.8571	12–14 (86%); 12,13,15 (10%); 13–15 (5%)
Rank modes 4–5	2	35	12.6%	1	4–5	1	4–5 (100%)
Rank modes 13–15	3	29	10.4%	0.8966	12–14	0.5	12–14 (50%); 13–15 (31%); 12,14,15 (15%)
Rank modes 7–8	2	21	7.6%	1	7–8	1	7–8 (100%)

Table S3 — Per-mode swap rates

Mode	Swap rate	Fail rate	Non-id. rate	n swapped	n failed
1	0	0	0	0	0
2	0	0	0	0	0
3	0.003597	0	0.003597	1	0
4	0.2698	0	0.2698	75	0
5	0.3777	0	0.3777	105	0
6	0.1835	0	0.1835	51	0
7	0.01079	0	0.01079	3	0
8	0.003597	0	0.003597	1	0
9	0.2698	0	0.2698	75	0
10	0.2698	0	0.2698	75	0
11	0.003597	0	0.003597	1	0
12	0.6151	0.007194	0.6223	171	2
13	0.6295	0	0.6295	175	0
14	0.5072	0.01439	0.5216	141	4
15	0.3813	0.1978	0.5791	106	55

Table S4 — Adjacent rank-pair swap diagnostics

cluster	cluster_key	rank_pair	ref_pair_mode	swap_rate	invalid_rate	match_rate	median_gap_in	near_deg_rate	true_deg_rate	peak_rate	active_frac	cv_swap	verdict
Rank modes 9-10	modes_9,10	9 $\Rightarrow$ 10	9 $\Rightarrow$ 10	0.3750	0.0000	1.0000	0.1206	0.5468	0.0000	1.0000	0.2050	0.6157	veering
Rank modes 12-15	modes_12,15	12 $\Rightarrow$ 13	12 $\Rightarrow$ 13	0.4154	0.0345	0.5603	0.0516	0.8597	0.0000	1.0000	0.3094	0.7588	veering
Rank modes 12-15	modes_12,15	13 $\Rightarrow$ 14	13 $\Rightarrow$ 14	0.1429	0.0603	0.3017	0.0802	0.7914	0.0000	1.0000	0.0324	2.0521	veering
Rank modes 12-15	modes_12,15	14 $\Rightarrow$ 15	14 $\Rightarrow$ 15	0.1818	0.2241	0.3793	0.0721	0.7734	0.0000	1.0000	0.0935	1.6757	veering
Rank modes 5-6	modes_5,6	5 $\Rightarrow$ 6	5 $\Rightarrow$ 6	0.1282	0.0000	0.8298	0.1661	0.3633	0.0000	1.0000	0.0288	1.6080	veering
Rank modes 12-13	modes_12,13	12 $\Rightarrow$ 13	13 $\Rightarrow$ 14	0.0811	0.0128	0.4744	0.0516	0.8597	0.0000	1.0000	0.0432	2.9996	veering
Rank modes 14-15	modes_14,15	14 $\Rightarrow$ 15	12 $\Rightarrow$ 15	0.6296	0.1429	0.3857	0.0721	0.7734	0.0000	1.0000	0.3921	0.8933	veering

Table S5 — Cluster-level exchange classification

<i>cluster</i>	<i>cluster_id</i>	<i>n</i>	<i>samplesize</i>	<i>mean_mutations_per_site</i>	<i>seq_type</i>	<i>swap_order</i>	<i>full_swap_order</i>	<i>ref_mutations_per_site</i>	<i>info_loss_ratio</i>	<i>avg_swap_rate</i>	<i>fold_increase</i>	<i>wilcoxon_z_score</i>	<i>t_test_p_value</i>	<i>cyclic_order</i>	<i>avg_swap_rate</i>	<i>avg_swap_rate</i>
Rank nodes 9-10	<i>nodes_9-10</i>	2	152	54.7%	0.0811	near-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	9-10	1.0000	1.0000	0.3750	0.7500	0.3750	0.6250	0.6250
Rank nodes 12-15	<i>nodes_12-15</i>	4	116	47.1%	0.0315	near-degenerate	VEERING (localised character exchange)	MIXING (multi-mode cycles, 2 common)	12-15	1.0000	0.7414	0.6431	0.2536	0.7414	0.6431	0.6431
Rank nodes 10-12	<i>nodes_10-12</i>	3	183	58.9%	0.0777	near-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	10-12	1.0000	0.8289	0.6250	0.3750	0.8289	0.6250	0.6250
Rank nodes 12-13	<i>nodes_12-13</i>	2	78	28.1%	0.0308	near-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	13-14	0.8405	0.9872	0.0385	0.0219	0.8744	0.9811	0.9811
Rank nodes 13-15	<i>nodes_13-15</i>	3	71	28.1%	0.0308	near-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	13-14	0.8405	0.9872	0.0385	0.0219	0.8744	0.9811	0.9811

Table S6 — Morphology-axis permutation null test

Cluster	Rank pair	Peak _{obs}	p (peak)	Active _{obs}	p (active)	CV _{obs}	p (CV)	Significant
Rank modes 9–10	9 $\rightleftharpoons$ 10	1	0.2547	0.2162	0.8202	0.6042	0.02697	NO
Rank modes 12–15	12 $\rightleftharpoons$ 13	1	0.4496	0.283	0.7223	0.5553	0.4046	NO
Rank modes 12–15	13 $\rightleftharpoons$ 14	0.25	0.8891	0.5	0.3437	1	0.4336	NO
Rank modes 12–15	14 $\rightleftharpoons$ 15	0.5	0.7443	0.09302	0.8492	0.9727	0.4256	NO
Rank modes 5–6	5 $\rightleftharpoons$ 6	0.5	0.9001	0.1267	0.4535	1.048	0.7263	NO
Rank modes 12–13	12 $\rightleftharpoons$ 13	0.25	0.8192	0.4375	0.2438	1.146	0.5455	NO
Rank modes 14–15	14 $\rightleftharpoons$ 15	0.75	0.6563	1	0.5504	0	1	NO

Supplementary Note 2 — Cohort robustness and sex effects

The main text uses sex effects only where they materially sharpen interpretation of the recurrent clustered subspaces, especially the 9–10 inlet-related pair. The figure and tables below provide the broader sex-stratified summary and the bootstrap evidence that the dominant clusters are stable features of the cohort rather than products of a few outlying subjects.

Table S7 — Bootstrap robustness under random 80% subcohorts

Metric	Target	Observed	Boot. mean	Boot. std	95% CI
prevalence	Rank modes 9–10	0.547	0.548	0.015	[0.518, 0.577]
prevalence	Rank modes 12–15	0.417	0.417	0.015	[0.387, 0.450]
prevalence	Rank modes 5–6	0.338	0.338	0.014	[0.311, 0.365]
prevalence	Rank modes 12–13	0.281	0.280	0.014	[0.252, 0.311]
prevalence	Rank modes 14–15	0.252	0.252	0.013	[0.225, 0.275]
swap_rate	Mode 4	0.270	0.270	0.014	[0.243, 0.295]
swap_rate	Mode 5	0.378	0.379	0.015	[0.351, 0.405]
swap_rate	Mode 6	0.183	0.183	0.012	[0.158, 0.207]
swap_rate	Mode 7	0.011	0.011	0.003	[0.005, 0.014]
swap_rate	Mode 9	0.270	0.270	0.013	[0.243, 0.293]
swap_rate	Mode 10	0.270	0.270	0.013	[0.243, 0.293]
swap_rate	Mode 12	0.594	0.593	0.014	[0.565, 0.622]
swap_rate	Mode 13	0.622	0.622	0.015	[0.592, 0.653]
swap_rate	Mode 14	0.468	0.467	0.015	[0.439, 0.500]
swap_rate	Mode 15	0.453	0.453	0.015	[0.423, 0.484]
median_gap	Pair 1–2	1.609	1.608	0.015	[1.572, 1.634]
median_gap	Pair 2–3	1.183	1.186	0.010	[1.169, 1.206]
median_gap	Pair 3–4	0.484	0.483	0.008	[0.465, 0.495]
median_gap	Pair 4–5	0.271	0.271	0.005	[0.261, 0.281]
median_gap	Pair 5–6	0.166	0.166	0.003	[0.156, 0.170]
median_gap	Pair 6–7	0.526	0.527	0.005	[0.516, 0.539]
median_gap	Pair 7–8	0.287	0.287	0.005	[0.277, 0.297]
median_gap	Pair 8–9	0.913	0.912	0.007	[0.895, 0.922]
median_gap	Pair 9–10	0.121	0.121	0.003	[0.117, 0.127]
median_gap	Pair 10–11	1.144	1.146	0.012	[1.129, 1.165]
median_gap	Pair 11–12	0.705	0.704	0.005	[0.694, 0.712]
median_gap	Pair 12–13	0.052	0.052	0.002	[0.049, 0.056]
median_gap	Pair 13–14	0.080	0.080	0.002	[0.076, 0.085]
median_gap	Pair 14–15	0.072	0.071	0.003	[0.065, 0.076]

Figure S1 — Sex-stratified routing distributions across all fifteen modes

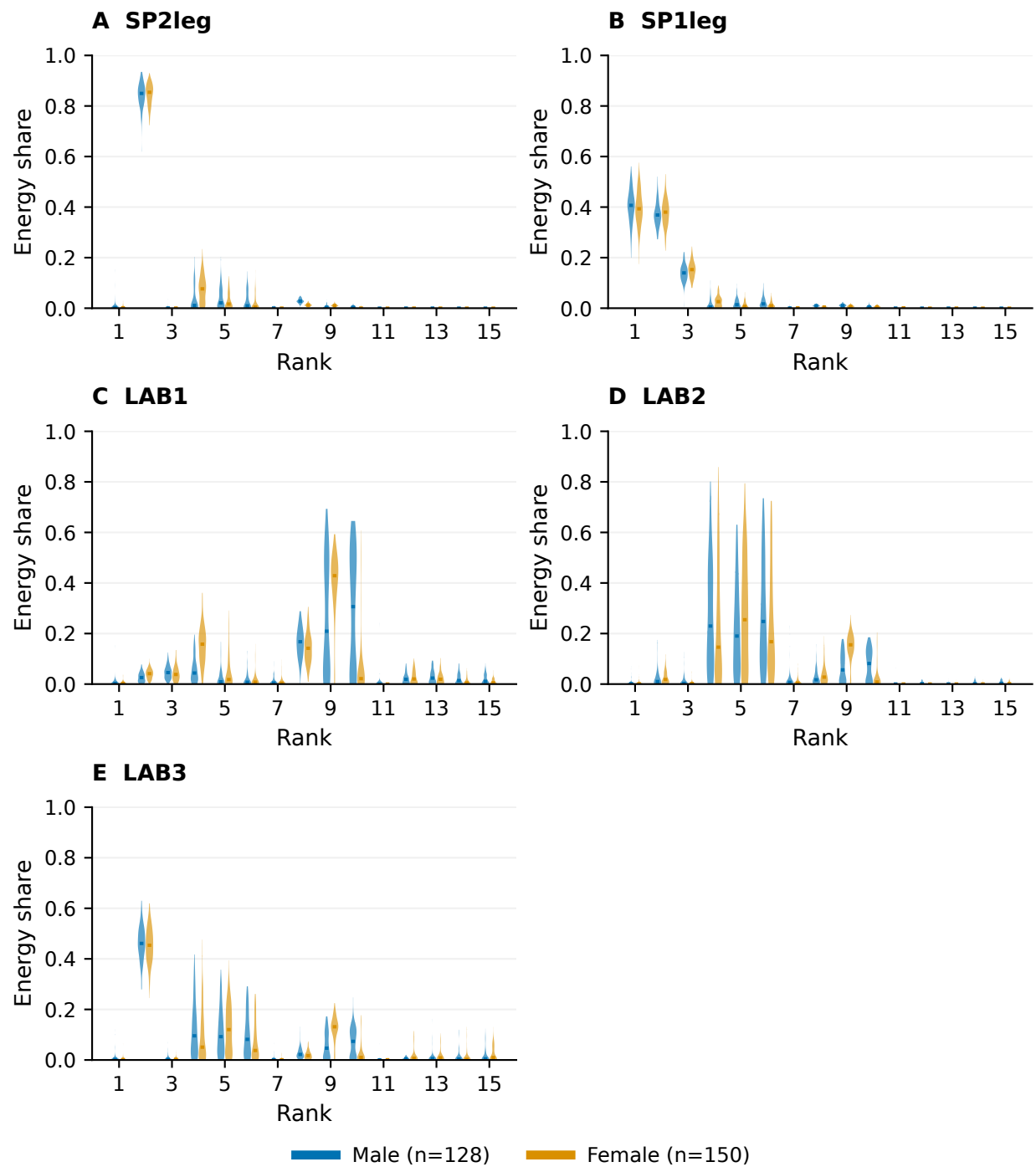


Table S8 — Sex-stratified tests for spectral and subspace metrics

Metric	Male med.	Female med.	U	p	r_{rb}
gap_in (9–10)	0.104	0.1534	6384	1.489e-06	0.335
gap_in (12–13)	0.04925	0.0552	9182	0.5321	0.04354
gap_in (13–14)	0.09191	0.07643	1.063e+04	0.1233	-0.1073
gap_in (14–15)	0.06002	0.08448	7631	0.003216	0.2051
gap_in (5–6)	0.1669	0.166	9677	0.9088	-0.008021
gap_in (4–5)	0.2627	0.2768	9299	0.6529	0.03135
gap_in (7–8)	0.3604	0.245	1.412e+04	1.311e-11	-0.471
swap_count_per_subject	4	3	1.352e+04	2.824e-09	-0.4079
permutation_rate	0.8984	0.9	0.9829	1	-0.01724
d_G (modes_9_10)	0.2605	0.3996	377	1.5e-19	0.8649
d_G (modes_12_15)	1.455	0.7866	2848	6.629e-11	-0.7059
d_G (modes_5_6)	0.8472	0.572	1793	1.766e-07	-0.6263
d_G (modes_12_13)	0.5299	0.3729	1120	0.0001815	-0.4973
d_G (modes_14_15)	0.5736	0.9485	270	6.922e-05	0.5559

Supplementary Note 3 — Load-routing specificity

The main manuscript uses standing and proxy childbirth loads to show the contrast between a robust low-rank backbone and selectively recruited higher-rank subspaces. The materials below provide the fuller load-specificity diagnostics, including cluster-level function-locking, dominant-mode sets, and pairwise overlap of full routing profiles.

Figure S2 — Load-function coupling correlations across prevalent clusters

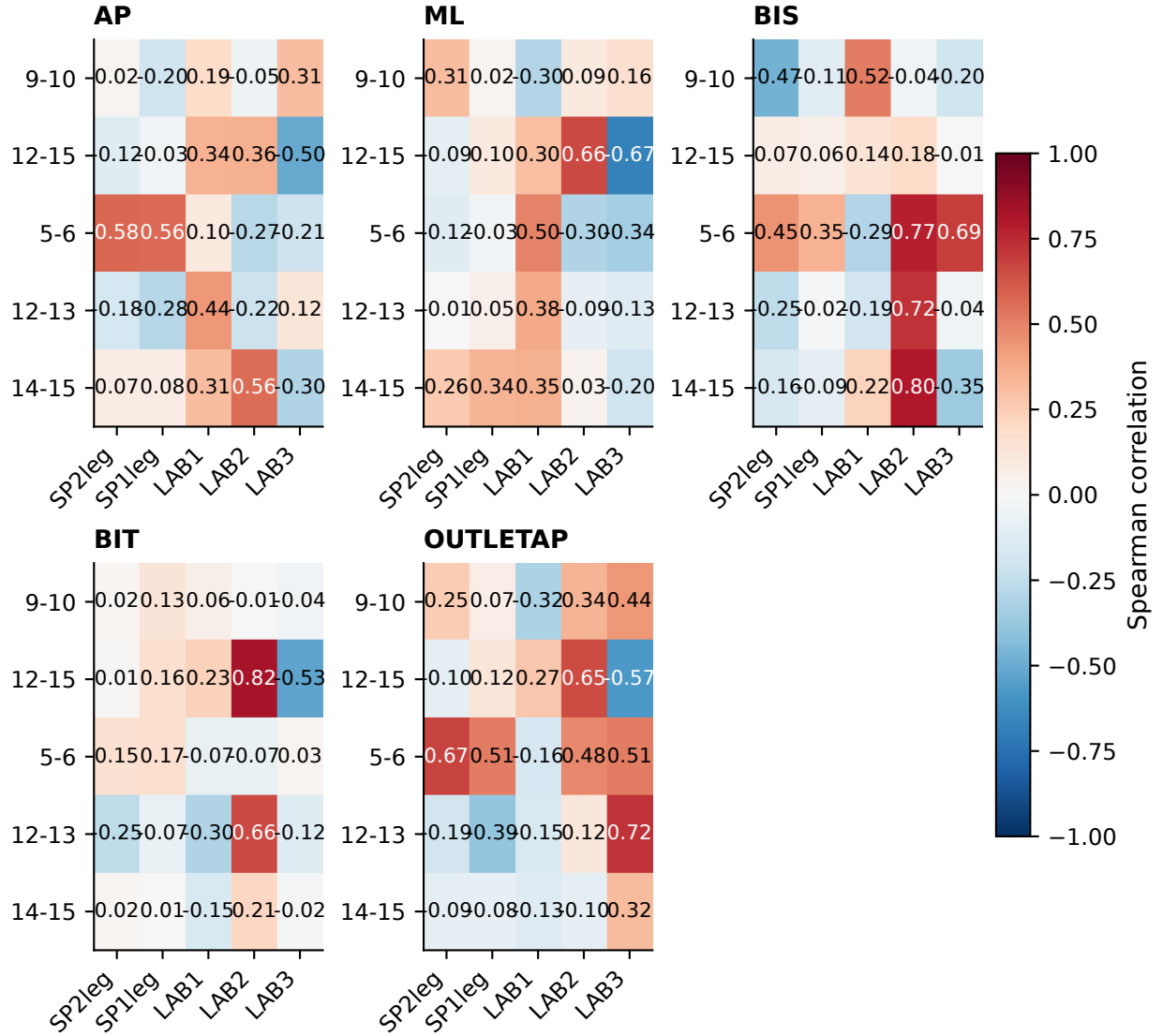


Table S9 — Load-function coupling synthesis

Cluster	Regime	Best-predictor coupling axis for FLI (AP/ML inlet, BIS/BIT outlet, or Outlet AP scalar)	FLI ρ	Load case	$\bar{f}_C$	$r_{rb}(dG)$
Rank modes 9–10	Veering	BIS outlet	0.5144	LAB ₁	0.5291	0.8649
Rank modes 9–10	Veering	Outlet AP scalar	0.3508	LAB ₂	0.1578	0.8649
Rank modes 9–10	Veering	Outlet AP scalar	0.4828	LAB ₃	0.1407	0.8649
Rank modes 12–15	Mixing (4D)	AP inlet	0.4858	LAB ₁	0.07568	-0.7059
Rank modes 12–15	Mixing (4D)	ML inlet	0.7	LAB ₂	0.01016	-0.7059
Rank modes 12–15	Mixing (4D)	ML inlet	-0.5942	LAB ₃	0.06857	-0.7059
Rank modes 5–6	Weak exchange	BIS outlet	-0.2788	LAB ₁	0.02502	-0.6263
Rank modes 5–6	Weak exchange	BIS outlet	0.6786	LAB ₂	0.5917	-0.6263
Rank modes 5–6	Weak exchange	BIS outlet	0.6589	LAB ₃	0.217	-0.6263
Rank modes 12–13	Weak exchange	BIT outlet	0.6247	LAB ₁	0.06863	-0.4973
Rank modes 12–13	Weak exchange	BIT outlet	-0.5387	LAB ₂	0.0009939	-0.4973
Rank modes 12–13	Weak exchange	Outlet AP scalar	0.7433	LAB ₃	0.01162	-0.4973
Rank modes 14–15	Mixing	AP inlet	0.3085	LAB ₁	0.02817	0.5559
Rank modes 14–15	Mixing	AP inlet	0.6098	LAB ₂	0.01237	0.5559
Rank modes 14–15	Mixing	BIT outlet	-0.54	LAB ₃	0.008386	0.5559

Table S10 — Dominant mode sets for standing and proxy loading

Load	Top mode	Top mode share (%)	Top-3 modes	Top-3 share (%)	N_{80}	Modes in 80% set	Cumulative share at N_{80} (%)
SP2leg	2 (ML rotation)	84.49	2,4,5	93.79	1	2	84.49
SP1leg	1 (CC rotation)	39.89	1,2,3	92.04	3	1,2,3	92.04
LAB ₁	9 (inlet opening)	33.49	9,10,8	66.12	5	9,10,8,4,3	81.48
LAB ₂	5	25.15	5,6,4	74.77	4	5,6,4,9	85.8
LAB ₃	2 (ML rotation)	45.81	2,5,4	68.09	5	2,5,4,9,6	85.8

Table S11 — Pairwise overlap of load-routing profiles

Load A	Load B	Jaccard overlap (80% sets)	Cosine similarity (full profile)	Shared modes (80% sets)
SP2leg	SP1leg	0.3333	0.6668	2
SP2leg	LAB ₁	0	0.1202	
SP2leg	LAB ₂	0	0.1343	
SP2leg	LAB ₃	0.2	0.9367	2
SP1leg	LAB ₁	0.1429	0.1233	3
SP1leg	LAB ₂	0	0.1051	
SP1leg	LAB ₃	0.1429	0.6326	2
LAB ₁	LAB ₂	0.2857	0.48	4,9
LAB ₁	LAB ₃	0.25	0.3613	4,9
LAB ₂	LAB ₃	0.8	0.4456	4,5,6,9

Supplementary Note 4 — Model-boundary, sensitivity, and deformation atlas

These analyses bound the main claims rather than extend them. Allometry quantifies what is lost when the main spectral analysis is performed on size-normalized pelvises. Material uncertainty quantifies how much absolute eigenvalue identifiability would degrade under broader soft-tissue and density-law scatter, while showing that the relative fragile-versus-robust gap architecture is preserved. Robustness checks confirm stability across percentile thresholds and reference subject choices. Finally, the full 15-mode displacement field atlas is rendered on the common reference domain.

Figure S3 — Robustness and reconstruction error

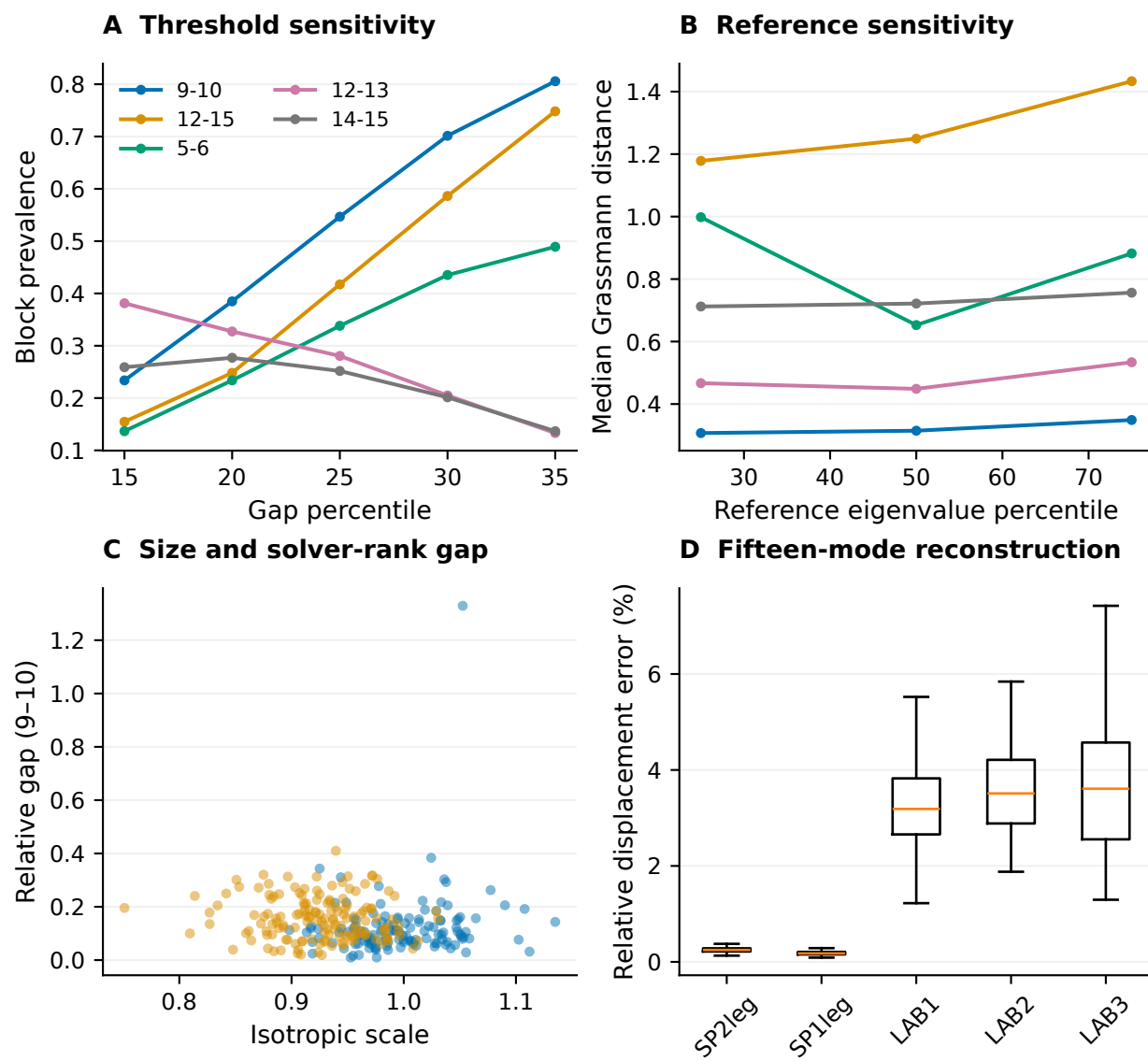


Figure S4 — Reference pelvic deformation atlas (fifteen modes)

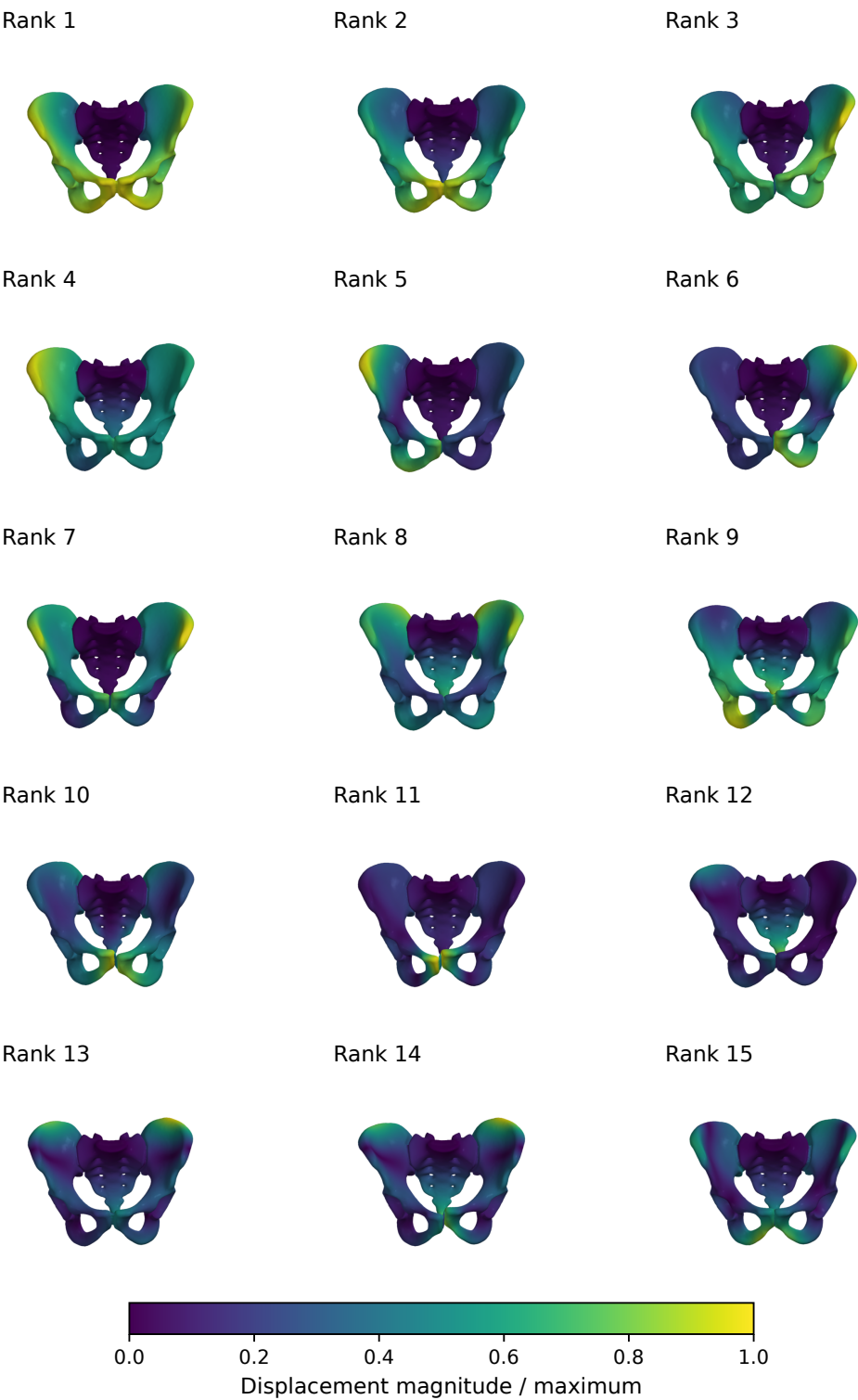


Figure S5 — Allometry dashboard

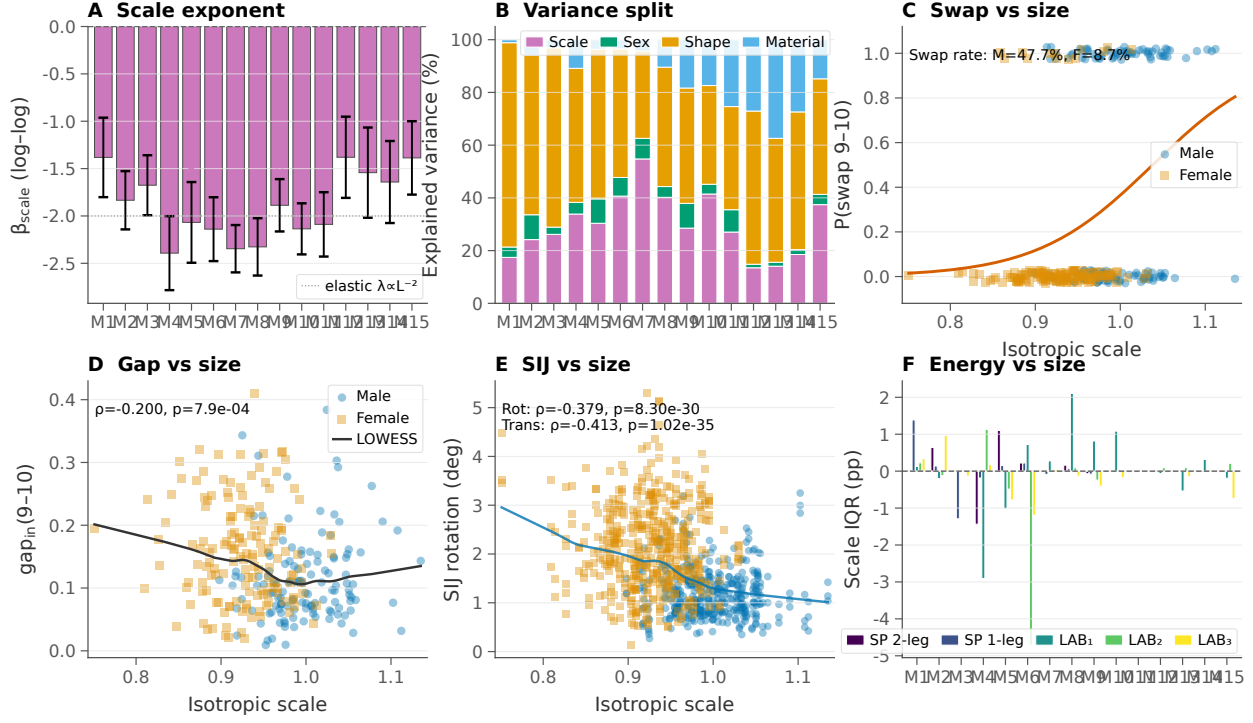


Table S12 — Allometry summary

Table 1: Allometric scaling exponents and variance decomposition per eigenvalue mode. β_{scale} is the IV-2SLS log-log coefficient of eigenvalue on isotropic scale, with HC-robust 95 % CIs and FDR-corrected p -values. $\Delta\lambda$ per IQR gives the percent eigenvalue change for an interquartile range shift in log-scale. R^2 columns show the LMG (Shapley) variance share for each predictor group.

Mode	β_{scale}	95% CI	p_{FDR}	$\Delta\lambda$ IQR (%)	R^2 scale (%)	R^2 shape (%)	R^2 material (%)	R^2 sex (%)
1	-1.38	[-1.80, -0.96]	4.7e-10	-10.1	17.4	77.5	1.1	3.9
2	-1.83	[-2.14, -1.53]	8.4e-26	-13.1	24.2	60.6	5.8	9.4
3	-1.68	[-1.99, -1.36]	2.0e-21	-12.1	26.2	68.0	3.1	2.7
4	-2.39	[-2.78, -2.00]	7.7e-27	-16.8	33.9	50.8	10.9	4.4
5	-2.07	[-2.49, -1.64]	1.1e-18	-14.7	30.4	56.8	3.6	9.3
6	-2.14	[-2.48, -1.80]	3.4e-28	-15.1	40.7	48.8	3.5	7.1
7	-2.35	[-2.60, -2.10]	5.7e-49	-16.5	54.8	33.1	4.3	7.8
8	-2.33	[-2.63, -2.02]	2.4e-37	-16.4	40.2	45.3	10.4	4.1
9	-1.89	[-2.16, -1.61]	1.9e-31	-13.5	28.5	43.8	18.3	9.4
10	-2.14	[-2.41, -1.87]	7.7e-39	-15.1	41.5	37.4	17.4	3.7
11	-2.09	[-2.43, -1.75]	6.3e-27	-14.8	27.0	39.1	25.4	8.5
12	-1.38	[-1.81, -0.95]	9.4e-10	-10.1	13.5	58.1	27.1	1.3
13	-1.54	[-2.02, -1.07]	8.2e-10	-11.2	14.1	47.0	37.4	1.5
14	-1.64	[-2.07, -1.21]	1.5e-12	-11.8	18.5	52.2	27.4	1.9
15	-1.39	[-1.77, -1.00]	1.8e-11	-10.1	37.4	43.8	14.8	3.9

Table S13 — Combined first-order material uncertainty summary

Table 2: Population-level material uncertainty propagation stratified by sex. CoV is the total coefficient of variation (ligament + cartilage + bone). Lig/Cart/Bone columns show each group’s fraction of total variance. Gap Z -score measures robustness of each spectral gap to combined material noise (first-order signal-to-noise ratio; not a confidence probability).

Mode	Female ($N = 150$)				Male ($N = 128$)			
	CoV (%)	Lig (%)	Cart (%)	Bone (%)	CoV (%)	Lig (%)	Cart (%)	Bone (%)
1	46.0	5	94	0	46.5	7	93	1
2	46.5	4	93	2	46.7	5	93	2
3	42.5	5	94	1	43.3	6	93	1
4	46.0	6	90	4	45.1	8	90	2
5	44.2	8	89	3	44.2	9	89	2
6	44.9	9	89	2	44.7	10	88	2
7	45.9	16	83	1	45.5	15	84	1
8	47.3	7	90	3	46.9	6	91	3
9	45.3	4	88	7	43.9	4	90	5
10	43.8	2	90	8	44.1	5	90	6
11	44.5	1	87	12	43.7	3	89	9
12	45.7	1	85	14	45.4	2	85	13
13	46.3	1	84	15	46.1	1	83	16
14	45.4	1	86	13	45.2	2	87	10
15	45.4	2	88	10	45.0	3	89	8